Reserve-based lenders must adapt risk framework as energy producers package best collateral in asset-backed securities

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A financing structure known as "proved developed producing asset-backed securities"—PDP ABS in industry vernacular—is becoming increasingly prevalent in today's energy landscape. But it may be unfamiliar to some in the traditional banking sector.

“Proved developed producing” denotes the highest-quality reserves in any exploration and production portfolio. A PDP ABS is a fixed-income security backed by cash flows from producing oil and gas reserves.

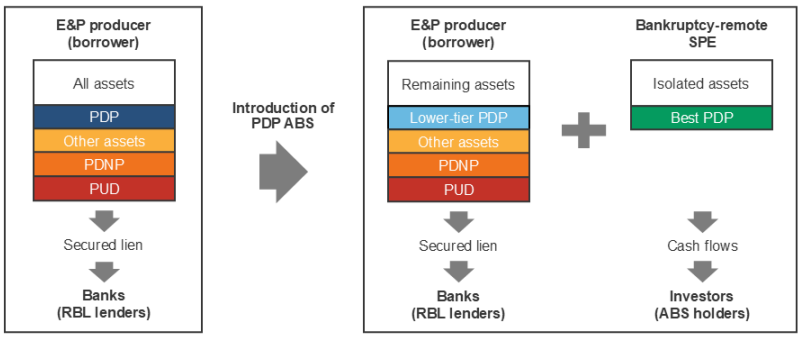
The PDP ABS market is still relatively small. Since its inception in 2019, cumulative issuance has reached approximately $20 billion across more than 15 issuers. That’s still a modest figure compared to the hundreds of billions in outstanding reserve-based lending (RBL) commitments. However, understanding this financing structure is now essential because it affects banks’ competitive positioning and portfolio risk management.

PDP ABS isolates producing assets in bankruptcy-remote structures

In traditional RBL, loans are secured by projected values of reserves. PDP ABS transfers specific producing assets into bankruptcy-remote, special-purpose entities (SPEs) that issue securities to institutional investors. These securities are repaid from predictable revenue generated by production held in SPEs.

While this new structure makes loans safer and more appealing to investors, it also means that once assets enter an SPE, they’re no longer available to meet a producer’s other obligations. In a bankruptcy scenario, these assets continue performing and paying ABS investors, while general creditors and RBL lenders must go through bankruptcy resolution. Chart 1 illustrates how PDP ABS shifts a borrower’s best-producing assets—best PDP—into an SPE, affecting RBL lenders’ collateral positions.

Chart 1
How PDP ABS changes lender collateral position



NOTES: In traditional reserve-based lending (RBL, left), banks hold secured claims on all producer assets. When producers issue proved developed producing asset-backed securities (PDP ABS, right), the highest-quality producing assets transfer to a bankruptcy-remote special-purpose entity (SPE). These isolated assets generate cash flows for institutional investors, while banks retain claims only on the remaining, higher-risk collateral. Asset colors indicate risk and quality levels: green represents best PDP reserves; light blue shows lower-quality PDP; dark blue indicates all PDP as an aggregate of best PDP and lower-quality PDP; yellow indicates other assets; orange shows proved developed non-producing (PDNP) reserves; red represents proved undeveloped (PUD) reserves—the highest-risk category. E&P is exploration and production.

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Why producers choose PDP ABS

PDP ABS addresses specific pain points that producers face with traditional bank financing.

First, it eliminates semiannual borrowing-base redeterminations. In traditional RBL, lenders review the value of underlying reserves twice yearly. These redeterminations recalculate how much capital borrowers can access based on current commodity prices and updated reserve value estimates.

When oil and gas prices fall, the borrowing base often shrinks, sometimes significantly. The volatile borrowing base creates uncertainty for borrowers, as their available credit can decrease precisely when market conditions become challenging. PDP ABS avoids this volatility through 10- to 15-year fixed amortization schedules that are not subject to periodic redetermination.

Second, economics favors the ABS structure. PDP ABS commonly offers 55–75 percent advance rates—the maximum amount borrowers can draw from lenders, based on a percentage of asset value—versus 40–50 percent in traditional RBL facilities.

The higher advance rates reflect the ABS structure's strict risk isolation and bankruptcy-remote features, which attract institutional investors like pension funds and insurance companies that typically would not invest in producers' lower-rated debt offerings.

For a producer with $500 million in high-quality PDP assets, traditional RBL might provide $200 million–$250 million in available capital (40–50 percent advance rate). PDP ABS could provide $275 million–$375 million (55–75 percent advance rate)—a difference of $75 million–$125 million in additional borrowing capacity.

Third, the structure provides fixed-rate financing that pairs naturally with mandatory commodity hedging, typically covering 75–85 percent of production.

Traditional RBL facilities commonly carry variable interest rates tied to benchmarks like the Secured Overnight Financing Rate (SOFR). They also have shorter commitment periods, typically 3–5 years, like corporate lines of credit.

Banks do not offer fixed-rate RBL tranches because the borrowing base fluctuates periodically, and funds are drawn and repaid flexibly.

By shifting from RBL to PDP ABS, borrowers accept strict hedging requirements and asset isolation in exchange for higher proceeds, longer certainty and elimination of redetermination volatility.

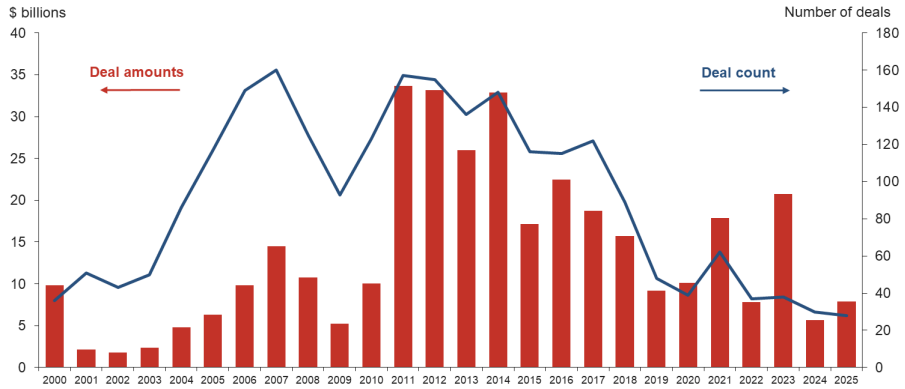
Capital-market shifts drive PDP ABS growth

Growing borrower interest in PDP ABS coincides with significant supply-side constraints reshaping traditional RBL availability.

Environmental, social and governance (ESG) commitments drove widespread bank retrenchment from oil and gas lending in the post-COVID period. Major European banks—historically significant energy lenders—substantially reduced their upstream energy portfolios or exited the sector entirely. Large U.S. banks followed similar paths, though some regional banks increased their energy exposure, partially filling the gap.

Simultaneously, private equity capital allocated to upstream energy has declined sharply. Chart 2 shows the decline in private equity (PE) volume in the upstream energy sector from 2011 to present. Annual PE deal volume has fallen from peaks of $30 billion–$35 billion in 2011–2014 to $5 billion–$10 billion in recent years, a reduction of more than 75 percent.

Chart 2
Private equity investment in upstream oil and gas down sharply from peak levels



NOTE: Deal amounts and deal counts for executed PE transactions in upstream oil and gas sector, 2000–2025. Peak PE activity of $30–35 billion annually (2011–2014) has declined by more than 75 percent to $5–10 billion in recent years, with deal counts falling from 140–160 to 30–40 annually.
SOURCE: S&P Capital IQ.

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These pressures create financing gaps precisely as producers need capital. PDP ABS fills this void by tapping insurance companies, pension funds and fixed-income asset managers, institutional investors that face fewer ESG constraints and actively seek long-duration, high-quality cash flows.

Producers use both structures strategically

PDP ABS is optimized for monetizing mature, stable production. RBL facilities remain better suited for funding ongoing development.

Increasingly, producers recognize they can use both. They securitize mature PDP assets to maximize proceeds while maintaining smaller RBL facilities to fund development.

This dual-structure approach creates intercreditor complexity. Typically, ABS facilities are senior to RBL in the capital structure, with SPE assets carved out entirely. Traditional lenders must underwrite to the remaining collateral pool while accepting that their liens are effectively subordinated to securitized assets.

Reserve-based lenders could face greater risk

The growth of PDP ABS creates interconnected challenges for traditional reserve-based lenders. While the market remains modest in absolute size, its impact on RBL portfolios is disproportionate.

The fundamental issue is a straightforward economics problem known as adverse selection. Loans to borrowers that accept banks' terms may be riskier than loans to borrowers that could access capital through other means. With stable cash flows, proven reserves and minimal execution risk, PDP assets represent the lowest-risk collateral in any exploration and production portfolio. When borrowers securitize these assets, traditional RBL lenders are left with claims on the riskier remaining assets.

That remainder includes proved undeveloped (PUD) reserves that require significant capital expenditure to bring to production. It includes proved developed nonproducing (PDNP) assets awaiting completion or workover. And it includes producing properties in less desirable basins with higher decline rates or operating costs.

PDP ABS adoption could hurt RBL profitability

Revenue impact compounds the credit quality deterioration. If borrowers can access higher loan-to-value ratios through ABS versus RBL, banks’ RBL facilities increasingly become backups used as revolver loans.

And as borrowers' usage of such revolver loans drops, capital committed to those loans cannot be reallocated to make new loans to offset the resulting drop in interest income. So, overall return on committed capital—total profitability relative to funds set aside for those loans—deteriorates precisely as their underlying risk increases.

RBL underwriting, oversight should evolve as PDP ABS grows

The remaining collateral requires fundamentally different oversight. PUD valuations are more sensitive to commodity price volatility. Development execution—the process of drilling wells, completing them for production and connecting them to processing facilities—becomes more central to credit quality. This process typically requires between $500,000 and $8 million–$15 million per well, depending on depth and completion techniques, like multistage hydraulic fracturing.

RBL traditionally involves covenants that bind borrowers to maintain certain financial metrics, such as debt-service coverage requirements or minimum current ratios.

Compliance with these covenants hinges on stable production revenue. When borrowers move their best-performing assets—and the revenue they generate—into SPEs for the purpose of issuing asset-backed securities, covenant compliance becomes more challenging.

This issue becomes particularly prevalent during development phases, when heavy spending on drilling puts added pressure on revenue from producing assets. As PDP ABS structures become more prevalent, banks may wish to revise covenant terms to ensure their claims on residual assets are adequately protected.

As PDP ABS becomes more common, it may be prudent to monitor competitive pressures' effects on RBL underwriting standards. Danger signals can include advance rates increasing without commensurate pricing adjustments; extended redetermination periods; or relaxed hedging requirements.

To date, we have not observed deterioration in RBL underwriting standards. However, banks should remain vigilant as the market evolves.

Attempting to compete directly with ABS economics without appropriate risk adjustments could expose lenders to precisely the risks that ABS structures are designed to avoid.

Adapting risk management practices

Several measures can help banks maintain underwriting discipline:

- Recalibrate advance rates and pricing to reflect higher risk profiles when PUD and PDNP assets dominate the borrowing base and apply more conservative loan-to-value ratios and risk-adjusted pricing to undeveloped reserves.
- Strengthen hedging requirements for borrowers that have securitized stable production, which creates heightened commodity price exposure during capital-intensive drilling programs.
- Enhance monitoring and covenant frameworks through more frequent oversight during development phases, real-time production data integration and tighter financial covenants that provide earlier warnings of problems.
- Require detailed development plans as part of the underwriting process and institute quarterly variance reporting to compare actual progress with planned timelines and budgets.

The path forward

PDP ABS is a financial innovation that has given energy producers access to new sources of institutional capital investment. Lowering the barriers to financing energy production creates opportunities for producers in this capital-intensive sector, and strengthening the energy sector can support continued economic growth more broadly.

For reserve-based lenders, this shift still requires careful attention. While the RBL market remains much larger than the ABS market, the role and risk profile of traditional RBL is evolving. Lenders should recognize this evolution and adapt their risk frameworks accordingly to avoid deteriorating portfolio quality and compressing returns. And bank examiners and supervisors must closely monitor the effects on RBL underwriting standards as this market evolves.

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